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Chapter 4: Fiber—The Possibility Space Between Observation and Truth: A Conceptual Analysis Framework

Abstract

Between any observation and the truth, there always lies a "possibility space"—the fiber: the set of all possible true states compatible with a reading. The size of a fiber must be interpreted under a specified state space and measure: under a given measure, the smaller the fiber, the more possibilities the reading rules out; a singleton fiber means the observation uniquely determines the truth; a fiber that is the full set means the reading cannot rule out any state. This chapter establishes the basic concepts and analytical representations of the fiber framework, introducing the conceptual partition of visible subspace and hidden subspace (relative to the knowledge state of a specified observer), as well as the concept of the "pseudo-small fiber"—a dangerous state in which the evidence-supported range of uncertainty remains large yet is judged by the observer to be small; preference-driven optimization and the observer's own cognitive biases may both foster it.

The fiber framework forms a structural isomorphism with the "Clock Room" thought experiment of Chapter 1, a conceptual complement to the Macro-Cognitive Accounting Framework (MCAF) of the previous chapter, and a structural isomorphism with Zhao Dingxin's "hermeneutic circle" theory. The value of the fiber framework lies not in eliminating fibers—for a non-singleton inverse image, the reading and its subsequent processing alone cannot eliminate the indistinguishability among those states—but in making fibers visible, comparable, and annotatable, thereby providing a public language for the AI engineering diagnostics of Part II and the financial

market analysis of Part III.

Keywords

Fiber; mapping function; uncertainty; epistemology; calibration; pseudo-small fiber; pseudo-calibration; visible subspace; hermeneutic circle

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Pre-reading Note: On the term "fiber"

This chapter will introduce a mathematical concept—"fiber." The reader need not be intimidated by this word. In this chapter, one only needs to grasp its intuitive core:

Fiber = the set of all possible true states compatible with a single reading.

For example: A circular shadow appears on a wall. Behind the wall there might be a sphere, a disk, or a cylinder. The shadow (the reading) does not tell the observer which object is actually behind the wall. The set of all objects that could cast this shadow is the "fiber" of that shadow.

This example will be fully developed in §1. This chapter uses "fiber" not because it is esoteric, but because it can precisely articulate something we have long felt but could not clearly state: **in observations involving information loss, the same reading may correspond to more than one true state.** The following translation table can serve as a quick reference for the entire chapter:

Fiber Language	Everyday Translation	Shadow Metaphor
Fiber	The set of all possible true states compatible with a reading	All objects that can cast the same shadow
Fiber is the full set	The reading is compatible with everything; it says nothing	This shadow could be caused by anything
Fiber is a singleton	The reading uniquely determines the true state	Only one kind of object can cast this shadow
Fiber shrinks	The range of possible true states narrows	Some possible objects have been ruled out
Fiber inflates	The range of possible true states widens; certainty is eroding	Objects previously thought ruled out have come back

One does not need to understand topology or fiber bundle theory; one only needs to carry this shadow metaphor while reading on.

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§1 Introduction: From Mathematics to Epistemology

"Fiber" is a precise concept in mathematics. Given a mapping $f : X \to Y$ from a set X to a set Y, for any $y \in Y$, the fiber of y is defined as $f^{-1}(y) = \{x \in X \mid f(x) = y\}$, i.e., the set of all elements x that

can be mapped to y. In topology, fiber bundle theory (Steenrod, 1951) reveals how the fiber bundle serves as a structure that is locally a direct product yet globally possibly nontrivial, describing the complex mapping relations between different spaces.

This chapter is not concerned with the topological properties of fibers. This chapter asks whether, when this mathematical concept is extracted from set theory and topology and placed in a completely different context—how humans come to know the truth through observation—it can provide a new language for **integrating** those **cognitive predicaments that have long been addressed separately by multiple disciplines but have lacked a unified intuitive framework**.

This predicament is: **in observations involving information loss, the same reading may correspond to more than one true state.**

A circular shadow appears on a wall. Behind the wall there might be a sphere, a disk, a cylinder, or an irregularly shaped object that, at a specific lighting angle, happens to cast a circular shadow. The shadow is an observed value. The set of all objects that could cast this shadow is the fiber of that shadow. The shadow itself does not tell the observer which object is behind the wall.

This is not a problem of intelligence—it is not that "if the observer were smarter, they could infer the object from the shadow." It is a structural problem: within the set of objects allowed in this example, different objects can produce the same circular silhouette, so this projection is not injective—the basis of indistinguishability is that different states yield the same reading, not merely that the number of dimensions differs before and after projection. The silhouette alone cannot uniquely determine the object behind the wall. Chapter 1 §3.5 has already independently argued this from the "Clock Room" thought experiment: when all mapping functions are unknown, all possible true states are compatible with the observations.

This intuition can be generalized to all observational activities. Any measuring instrument, any perceptual organ, any data collection process is a mapping function ($\pi : \mathcal{T} \to \mathcal{O}$) that maps the true state space ($\mathcal{T}$) to the observable space ($\mathcal{O}$). Given a reading $r \in \mathcal{O}$, the fiber $\pi^{-1}(r)$ is the set of all true states that could produce this reading. The size of the fiber—its "volume" or "dimension"—characterizes the range of possibilities the reading rules out: under a specified state space and measure, the smaller the fiber, the more possibilities the reading rules out; if the fiber is the full set, the reading cannot rule out any state.

This insight is not entirely new. In measurement theory, the representational theory established by Krantz et al. (1971) understands measurement as the construction of homomorphisms from empirical relational structures to numerical relational structures and requires clarification of the assumptions under which this construction holds. This chapter further connects that relational-structure perspective with instrument calibration: interpreting readings requires specifying the observation mapping and its conditions of applicability. In econometrics, the "partial identification" method developed by Manski (2003) argues that when the distribution of observed variables, together with the adopted assumptions, cannot uniquely determine the true value of the parameters, the compatible parameter values form an identification region—the identification region is not a singleton, and in the absence of effective restrictions it may be the entire parameter space—this can be regarded as a specialized counterpart of the fiber concept within a particular statistical model. In information theory, Shannon's (1948) conditional entropy $H(X|Y = y)$ measures the uncertainty still remaining after a given observation—this is one measure of fiber size in the information-theoretic sense. The Information Bottleneck method proposed by Tishby et al. (1999) further applies information-theoretic measures to the trade-off between compressing the

input and retaining relevant information, arguing that the optimal intermediate representation should retain information relevant to the output while discarding the rest—but the fiber framework presses one question further: do the dimensions discarded because they were judged "irrelevant" contain information crucial to certain groups or scenarios (see Chapter 13 §8 and Chapter 4 Appendix A §6.2, discussed later). In epistemology, the Duhem-Quine problem points out that empirical testing typically involves both theory and auxiliary hypotheses, so observational results may not adjudicate a single theoretical proposition on their own—the same observational evidence can be compatible with different theories (Quine, 1951).

In recent years, several exploratory attempts have emerged that introduce the fiber concept into different disciplines (the following two works are a preprint and a journal article respectively; their academic influence remains to be seen; they are cited here respectively as examples of the cross-domain diffusion of the fiber concept and of the same-word-different-meaning usage). In the philosophy of physics, Bamonti (2024) uses fiber bundle formalization to analyze the dependence of observables on reference frames in general relativity—the same model can yield different gauge-invariant readings under different reference frames; in the language of fiber bundles, selecting a reference frame amounts to introducing a section of the bundle. In AI evaluation, Bouattoura (2025)[1] proposes "fibristic epistemology" and argues that existing AI evaluation narratives frequently overestimate capacities and convey alternative realities rather than empirical truths. The paper also develops a logic-weighted simulation and rule-based scoring framework for AI tool evaluation, implemented as 114 simulated user entries with rule-based bonuses and penalties on top of base scores. In the field of machine learning, conformal prediction developed by Vovk et al. (2005) and Angelopoulos and Bates (2023) provides an engineering interface for uncertainty annotation: under applicable exchangeability and calibration conditions, a prediction set with a prespecified marginal coverage guarantee can be constructed without specifying a parametric form for the data distribution. The size of the prediction set can be used to evaluate the procedure's expression of uncertainty and its efficiency, but it does not directly equal the size of the inverse image of the observation mapping.

These works are each bound to specific domains—gauge invariance in physics, methodological procedures in AI evaluation, uncertainty quantification in machine learning—using different terminological systems. The fiber framework of this chapter is different: it is not a rigorous mathematical model or a domain-specific meta-methodology, but a cross-domain analytical tool oriented toward uncertainty management and cognitive debt accountability. Its originality lies not in inventing the insight of "fiber as an epistemological tool"—multiple academic traditions have independently developed similar ideas—but in providing a unified intuitive language and a cross-domain diagnostic system for the insights scattered across measurement theory, information theory, econometrics, and hermeneutic circle theory by constructing operational concepts such as "pseudo-small fiber," "gradient of detectability," and "perceived fiber."

The core propositions of the fiber framework can be summarized in three statements:

Proposition 1 (Information deficit renders the fiber ineliminable): For a given mapping, non-injectivity means that at least one reading corresponds to multiple true states; a particular reading fails to determine the true state uniquely only when its inverse image contains multiple states. When no constraint on the candidate mappings suffices to rule out any true state, the knowledge-conditioned identification set is the full set; this does not mean that the inverse image of the fixed actual mapping is itself the full set. In the absence of prior knowledge about the mapping function, observation cannot uniquely determine the truth.

[1]This chapter cites Bouattoura (2025) to compare a homonymous use of "fibristic" in AI evaluation and the paper's requirements for transparency and traceability, not to use the journal's standing as an authoritative basis for this book's framework.

Proposition 2 (Calibration narrows the identification set): Calibration can narrow the identification set by adding effective knowledge about the mapping function; for a fixed actual mapping and a fixed reading, the inverse image itself does not change as knowledge increases. The essence of calibration is the acquisition of information about the mapping function—whether through independent external reference ("opening a window to look at the sun") or through cross-validation among multiple mappings ("using multiple clocks to calibrate each other").

However, the effectiveness of the latter cannot be judged from agreement among outputs alone—if all mappings share the same systematic bias, the internal consistency shown by cross-validation cannot guarantee alignment with the truth. Cross-validation here serves the calibration function defined in Chapter 1 only when it introduces effective constraints that are independent of the loop of readings being calibrated and are in causal contact with the true state; comparisons confined to that loop cannot replace this condition.

Proposition 3 (Lost information cannot be recovered through post-processing alone): When a given mapping sends multiple true states to the same reading, that reading and its subsequent processing alone cannot guarantee unique recovery of the original true state. A reduction in dimension or coordinate count is not by itself sufficient to establish information loss; the state support and whether the mapping preserves the information needed to distinguish the relevant states must also be examined.

The data processing inequality expresses this post-processing limitation in information-theoretic terms: for a Markov chain $X \to Y \to Z$, $I(X;Z) \leq I(X;Y)$. Equality is allowed. The inequality states that subsequent processing cannot increase mutual information about X without introducing additional information about X; it does not by itself establish that a particular compression step is strictly lossy.

The relationship among the three propositions is as follows: Proposition 1 distinguishes the indistinguishability of observations from insufficient knowledge of the mapping; Proposition 2 concerns the conditions under which calibration narrows the identification set; Proposition 3 concerns the fact that, under a fixed lossy mapping, information already lost cannot be recovered by post-processing of the same reading—even if the calibration of Proposition 2 is completed, the loss described by Proposition 3 is still unrecoverable.

Yet none of the three propositions touches on the most dangerous situation. The most dangerous situation is not that the fiber is large—but that the fiber appears very small when it is actually very large. A large fiber that honestly admits uncertainty at least prompts the observer to initiate auditing; a genuinely small fiber itself rules out most candidate states. What truly breaks the observation-auditing loop is a system's rendering of a large fiber as if it were small on the visible subspace, so that the observer acquires a perception of certainty while the substantive uncertainty is preserved or even amplified. This situation will be formalized in §2.4 as the "pseudo-small fiber," which is the core diagnostic object repeatedly returned to throughout the rest of this chapter.

The structure of this chapter is as follows: §2 unfolds the basic definitions and analytical representations of the fiber framework, including the visible/hidden subspace decomposition, ordinal measures of fiber size, and the formal definition of the "pseudo-small fiber." §3 discusses the structural isomorphism between the fiber framework and the "Clock Room" thought experiment. §4 discusses the conceptual complementarity between the fiber framework and the MCAF framework. §5 discusses the resonance and divergence between the fiber framework and Zhao Dingxin's "hermeneutic circle" theory. §6 discusses the boundaries of the fiber framework—under what conditions it applies and under what conditions it does not. §7 is the conclusion.

§2 Formalization of the Fiber Framework

§2.1 Basic Definitions

Definition 1 (True State Space): There exists a true state space ($\mathcal{T}$), whose elements $T \in \mathcal{T}$ represent the complete true state of the object being observed. The true state space can have arbitrary dimensions and can be continuous or discrete. To the observer, the true state is not directly observable.[2]

Definition 2 (Mapping Function): The observation process is represented by a mapping function ($\pi : \mathcal{T} \to \mathcal{O}$), where the observable space ($\mathcal{O}$) is the space accessible to the observer. The mapping function encodes all characteristics of the observation process—the precision of the instrument, the limits of perception, the manner in which data is processed, the filtering effects of theoretical presuppositions. What the observer sees is only the reading $r = \pi(T)$.

Definition 3 (Fiber): Given an observed value $r \in \mathcal{O}$, the fiber of that observed value is defined as $\pi^{-1}(r) = \{T' \in \mathcal{T} \mid \pi(T') = r\}$. The fiber is a subset of the true state space—the set of all true states that could produce the observed value r.

The "size" of the fiber can be measured in multiple ways. If the true state space is a finite set, fiber size can be measured by its cardinality $|\pi^{-1}(r)|$. If the true state space is continuous, fiber size can be measured by its measure or topological dimension. If the state variable is discrete and a probability distribution is specified, the conditional entropy $H(T|R = r)$ can describe the uncertainty remaining under that reading; it is a distribution-weighted quantity, not a synonym for the cardinality of the inverse image. For continuous states, an applicable probability model and uncertainty measure must be specified separately; the full interpretation of discrete entropy cannot be carried over directly.

The fiber of a fixed mapping is an objectively defined inverse-image set. Its membership is determined by the mapping and the reading; its measure, topological dimension, or entropy also depends on the specified measure, topology, or probability distribution, and is not determined by the mapping structure alone. In practical applications, the mapping function itself may be partially unknown; in such cases, what we discuss is the "identifiable fiber"—the set of true states compatible with the observation under known constraints on the mapping function. This bears a structural correspondence to the "identification region" defined by Manski (2003) in econometrics, but the two are not equivalent in all settings. This chapter distinguishes the inverse image of the fixed actual mapping from the knowledge-conditioned identification set. Let Π_K be the family of candidate mappings admitted under knowledge condition K. The set of states compatible with reading r (namely the identifiable fiber) is then $\mathcal{I}_K(r) = \bigcup_{\tilde{\pi} \in \Pi_K} \tilde{\pi}^{-1}(r)$. When the actual mapping belongs to Π_K, this set contains the actual inverse image $\pi^{-1}(r)$. Calibration knowledge can make the identification set no larger by ruling out candidate mappings, but does not thereby change the inverse image of the fixed actual mapping; additional observations or changes to the observation apparatus require an explicit joint mapping or a new mapping.

Framework Positioning Statement. The above formal definitions constitute a **conceptual scaffold**, not a

[2]This framework presupposes a realist stance, i.e., that an object state exists independently of the current observation, and does not argue for this presupposition here. Other epistemological positions understand this premise and its scope of application differently; this does not warrant lumping social constructivism or hermeneutics as positions that reject the modeling of object states. The framework does not provide a defense for this presupposition but offers diagnostic tools for analysts who hold this presupposition.

computational procedure. In the vast majority of epistemological scenarios, the structure of $\mathcal{T}$ is unknown, and the absolute size of the fiber cannot be calculated. Therefore, the output of the fiber framework is typically a **qualitative risk annotation** (e.g., "high/medium/low pseudo-small fiber risk") or an **ordinal judgment** ("the fiber of A is larger than the fiber of B"). Its core value lies in cultivating a **habit of questioning**: when facing any observation, the first response is to assess "how large is the possibility space behind this reading?" The formal tools of the subsequent chapters should be understood within this positioning.

§2.2 Visible Subspace and Hidden Subspace

In many real-world scenarios, the observer does not completely lack knowledge of the mapping function, but possesses **partial** knowledge about it. The "observer" here is relational: it can be an individual, a research team, a regulatory body, or a disciplinary community—"observer" refers to the information receiver within a specific analytical context, whose knowledge state determines the boundary between visible and hidden. Different observers can have different visible/hidden partitions for the same system—this is not a defect of the framework, but a feature it actively describes. This leads to a key decomposition.

Definition 4 (Gradient of Detectability): The observer's degree of knowledge about the mapping function is not binary "know" or "do not know," but exists along a gradient of detectability—some dimensions are easier to calibrate than others, some deviations are easier to measure, some uncertainties are easier to perceive. As an analytical tool, we can conceptually partition the true state space into the visible subspace ($\mathcal{T}_v$, where the observer possesses relatively sufficient knowledge and the fiber can be constrained to a sufficiently small size within this subspace) and the hidden subspace ($\mathcal{T}_h$, where the observer lacks effective discriminative capacity and the fiber may be large or even the full set within this subspace), but it must be clear that this is not an absolute dichotomy, but rather the two ends of a continuous spectrum. In this chapter, for brevity of expression, the subsequent discussion will primarily use the binary notation $\mathcal{T}_v$ / $\mathcal{T}_h$, but the reader should understand them as "approximate classifications at the two ends of the detectability spectrum." A more complete formalized version would introduce a state-detectability scoring function $g : \mathcal{T} \to [0, 1]$, assigning different true states detectability scores relative to a specified observer and evidence conditions; describing detectability across different dimensions would additionally require a dimension index and its scoring function. This refinement lies beyond the scope of this chapter, but is a possible direction for future evolution.

Under this conceptual partition, when the reward objective genuinely covers the specified visible performance and the optimization effectively improves that performance, the system can fit the evaluation requirements better on the visible subspace—those specified visible behaviors better satisfy the evaluation criteria used by the reward. However, for the designated reference observer, there is currently insufficient evidence to confirm which effective constraints, if any, the hidden dimensions are subject to, or whether the system output therefore approximates the corresponding evaluation requirements. The optimization process may produce indirect constraints through model capacity limits, architectural regularization, etc., but for an observer lacking discriminative capacity, whether these constraints take effect, and in which direction they shape behavior, cannot yet be established from the observer's current evidence.

This reveals a counter-intuitive dynamic: **fixing visible problems may increase the relative weight of hidden deviations.** When the engineering team reduces failures in the visible subspace, this improvement alone does not determine how hidden deviations have changed. If hidden risk remains unchanged while visible risk falls, the proportion of hidden risk in the remaining risk rises; this is not an increase in its absolute amount or density.

Hidden behavior may still change through parameter coupling and other constraints, and the direction of change requires independent testing. And because the hidden subspace is precisely the set of dimensions that the observer cannot effectively discriminate, the corresponding deviation may not be detected by this observer in time. If visible improvement is narratively packaged as overall reliability, it may obscure hidden risks that remain unverified; this is related to the masking function of the narrative load (ϕ) defined in the previous chapter, but is not its sole mechanism, and new deviations are not ruled out.

Anchoring Statement: When this chapter and subsequent chapters use "visible subspace $\mathcal{T}_v$" and "hidden subspace $\mathcal{T}_h$," a **reference observer** must be specified, whether explicitly or implicitly. When not explicitly specified, the default reference observer is **the primary information receiver of the system** (for example: for an AI system, the end-user; for financial markets, market participants; for scientific measurement, the user of the measurement results). When the analysis requires switching the reference observer (e.g., switching from an ordinary user to a regulatory body), the switch and its effect on the boundary between $\mathcal{T}_v$ and $\mathcal{T}_h$ must be explicitly noted. This anchoring limits the range of constraints the observer can verify: for the specified observer, it remains unconfirmed which constraints actually operate in $\mathcal{T}_h$. Any further claim that the optimization process received no effective calibration signal for these dimensions requires separate evidence about the training data, reward function, and other sources of constraint.

§2.3 Ordinal Comparison of Fiber Size

The absolute size of the fiber is usually impossible to measure precisely—this requires complete knowledge of the mapping function and the structure of the true state space, which is typically unavailable in the open epistemic and socio-technical scenarios this framework primarily targets. But the true value of the fiber framework lies not in cardinal measurement, but in **ordinal comparison**: in many practical scenarios, the analyst can, based on partial knowledge of the mapping function, make well-grounded ordinal comparisons—judging whether fiber A is larger than fiber B—though such comparisons may yield disagreements due to differences in the analysts' knowledge.

When absolute size is unobtainable, some scenarios still admit well-grounded comparisons; where no set-inclusion relation or common measure exists, the objects should be marked as not yet comparable rather than forcibly ordered. When the mapping chain lengthens ($\pi = \pi_n \circ \cdots \circ \pi_1$), a set-inclusion comparison is possible only after the inverse images of each layer's readings are all pulled back to the same original state space; along a deterministic post-processing chain, the set of compatible states does not shrink, but need not strictly expand. The data processing inequality constrains average information; it does not directly compare single-layer fibers defined in different spaces. When an independent calibration source is introduced, the knowledge-conditioned identification set can narrow in the vicinity of the corresponding calibration point. The applicability of calibration constraints in extrapolation regions must be tested separately; "over-extrapolation" alone does not establish that the actual inverse image expands. When multiple independent mappings produce consistent outputs for the same truth and shared systematic bias has been ruled out, the knowledge-conditioned identification set can narrow; if they share a systematic bias, agreement alone does not establish narrowing—internal consistency is not equivalent to calibration.

An Illustration on a Historical Scale. Monetary history provides a case for comparing how mapping chains and calibration institutions change together, but chain length alone does not determine fiber size. Under the gold standard, banknotes nominally anchored to a fixed quantity of gold; the legally promised mapping chain

was short, and ordinary note-holders had a clear understanding of the conversion relationship—although the actual conversion capacity was constrained by the bank's reserve ratio, and a run could expose this hidden fiber. The Bretton Woods system lengthened the mapping chain: the franc was pegged to the dollar, the dollar pegged to gold. Most currency holders understood convertibility through monetary and institutional intermediaries; whether this increased uncertainty must be assessed after specifying convertibility entitlements, the target state, and institutional constraints. After the gold window closed in 1971, the ultimate calibration point disappeared—the previous commodity-convertibility constraint no longer applies, and compatible states and uncertainty about monetary value must be reanalyzed under the new institutional constraints—currency value was no longer defined by a single physical conversion ratio. This is not to say that modern fiat currency has lost its anchor—inflation targeting, central bank independence, and institutional credibility can serve different anchoring functions, whose effects must be compared with respect to specific objectives and institutional performance—but rather that the nature of the anchor shifted from a physical reference to an institutional commitment. The value of the fiber framework here is not to evaluate the merits of the gold standard versus fiat currency, but to press the question: under an institutional commitment as anchor, is the calibration point still clear, independent, and verifiable?

Boundary Conditions for Monotonicity: Along a Markov processing chain that introduces no additional information, mutual information about the original state cannot increase; for discrete variables with finite relevant entropies, average conditional entropy cannot decrease. This does not guarantee an increase in conditional entropy for every particular reading. Set comparisons must concern inverse images in the same original state space, and strict enlargement additionally requires a step to merge states that were previously distinguishable.

A distinction must also be drawn in practice between information preservation and the utilization capacity of a bounded observer: human language, as an intermediate representation, may in certain tasks make relevant structure easier to identify and use—for example, the linguistic statement 'the cup is on the table' more directly encodes the spatial relation of objects than the raw pixel matrix, even though it discards pixel-level detail. This can improve the task performance of a bounded observer, but it is not a counterexample to the claim that post-processing without additional information cannot increase task-relevant mutual information; if the abstraction process introduces external knowledge, that knowledge must be listed explicitly as an additional input.

Therefore, when the framework makes diagnoses based on the length of the mapping chain in subsequent chapters, two analytical purposes must be distinguished: **general-purpose diagnosis** ("What is the overall uncertainty structure of this system?")—mapping chain length can serve as a risk lead awaiting verification, its validity depending on the information each layer actually discards, the additional information introduced, and the calibration conditions; **task-specific diagnosis** ("How reliable is this system on task X?")—one must additionally assess whether each mapping layer preserves the information required to accomplish task X; where causal judgments are involved, the preservation of causal relations must be tested separately. The current framework primarily serves general-purpose diagnosis.

A Parallel Taxonomy from Game Design. The game design industry provides an already-engineered parallel taxonomy that classifies uncertainty by its sources. Greg Costikyan's (2013) *Uncertainty in Games*, published by MIT Press in the "Playful Thinking" series, opens with a core proposition:

> In the course of this book, I shall endeavor to persuade you that games require uncertainty to hold our interest, and that the struggle to master uncertainty is central to the appeal of games. (Costikyan,

2013)

The book argues that the core aesthetic of games is precisely the player's engagement with various sources of uncertainty in a protected environment. To make this aesthetic designable, Costikyan decomposes the uncertainty in games by its sources into eleven types: performative uncertainty (operational precision in action games), solver's uncertainty (whether a puzzle can be cracked), player unpredictability (what other players will do), randomness (dice and card draws), analytic complexity (combinatorial explosion in chess), hidden information (card games), narrative anticipation (how the story will unfold), development anticipation (postrelease content updates), schedule uncertainty (timed reengagement mechanics in social games), uncertainty of perception (picking out targets from a busy screen), and semiotic contingency (the unpredictability of the cultural meaning of outcomes). The book explains through concrete game mechanics how these sources affect uncertainty; some can be adjusted through design, but the classification itself does not prove that the sources are mutually independent.

The structural relationship between this taxonomy and the fiber framework is: Costikyan's eleven source-based types can be retrospectively reread by this chapter as different factors shaping the player's uncertainty structure, some adjustable through mechanism design **[T]**—but the game designer's perspective offers a reverse angle: in games, many types of uncertainty can be deliberately shaped by product mechanics; the size of the fiber faced by the player, which parts belong to the visible subspace, and which belong to the hidden subspace—these are influenced not only by design mechanics but also by player knowledge, interactive processes, and cultural interpretation outside the game, and are not all intentional settings by the designer. This provides two points of reference value for this chapter's framework. First, Costikyan's practice of distinguishing uncertainty by source provides a design vocabulary for comparing specific mechanisms, but it does not prove that the eleven sources can be independently added and subtracted, nor does it constitute a mathematical proof of this chapter's ordinal principle. Second, several years before this book's work on "uncertainty classification" in AI and financial contexts, it had already provided parallel evidence from a completely independent source: in the design practice of the entertainment industry, the systematic classification of uncertainty had already been codified in a tutorial textbook. Costikyan himself also clearly points out, in discussing solved games, that when the relevant uncertainty has been resolved for a particular player—for example, when the player has mastered the relevant strategy in a solved game—the game may lose its appeal; this is consistent with the proposition advanced at the outset of the book. In this chapter's translation, if relevant uncertainty is measured by discrete entropy, its value can approach zero; if measured by the cardinality of non-empty inverse images, the uniquely determinable state corresponds to a singleton rather than an empty set. This comparison shows that the uncertainty in gameplay must be discussed separately by object and by player knowledge; it is not an independent mathematical proof of the pseudo-small fiber mechanism.

§2.4 The Pseudo-Small Fiber

One of the most practically valuable concepts of the fiber framework is the **pseudo-small fiber**.

Definition 5 (Pseudo-Small Fiber): Let the true uncertainty range in the diagnostic situation be F_{true}—the set of true states compatible with reading r: when the mapping function is known, this is the actual inverse image $\pi^{-1}(r)$; when the mapping is partly unknown, it is characterized by the identification set $\mathcal{J}_K(r)$ defined in §2.1; this set contains the actual inverse image provided that the actual mapping belongs to the candidate family Π_K. Let the **perceived fiber** formed by the observer based on output content and presentation form

be $F_{perceived}$. When F_{true} is large but the perceived fiber is small, this state is called a **pseudo-small fiber**. The size of F_{true} must be assessed under explicit evidence conditions and comparison standards. It must be clarified: the "perceived fiber" $F_{perceived}$ is not another set, but the observer's cognitive representation of the fiber's size. We retain the term "perceived fiber" to maintain expressive consistency in the contrast between "true fiber vs. perceived fiber," but the reader must be clear: $F_{perceived}$ is not an inverse image under π, but a cognitive object constructed by the observer.

The pseudo-small fiber must be clearly distinguished from two other states: first, the **genuinely small fiber**, where true uncertainty is low and the observer's perception is aligned with it (F_{true} and $F_{perceived}$ are both small and aligned); this state does not by itself guarantee that every answer reported by the system is correct. Second, the **genuinely large fiber**, where the system honestly admits uncertainty, and the output on the visible subspace truthfully marks the uncertainty (F_{true} and $F_{perceived}$ are both large). The core feature of the pseudo-small fiber thus emerges: it is a systematic disconnection between an appearance of certainty ($F_{perceived}$ is small) and a substance of uncertainty (F_{true} is large). Under this ternary distinction, the genuinely small fiber does not rule out errors of answer selection or expression and remains open to external verification; the uncertainty label of the genuinely large fiber can prompt auditing. The risk of the pseudo-small fiber lies in the possibility that its appearance of certainty may lower the observer's willingness to initiate auditing, though external auditing may still identify it. These three categories serve to highlight the underestimation of uncertainty; they are not an exhaustive classification, and cases where uncertainty is overestimated are not the focus of this section.

The danger of the pseudo-small fiber lies in the fact that it may weaken the triggering conditions for the observer to initiate auditing. When a system outputs information in a definitive tone, fluent format, and confident appearance, the observer naturally forms the perception of a small fiber. But if the system's true fiber is large—it actually does not know the answer, or its answer has serious deviations in the hidden subspace—the observer may overlook the need for independent verification.

The key difference between the pseudo-small fiber and the genuinely large fiber that honestly discloses uncertainty is: the genuinely large fiber at least honestly annotates the uncertainty—when a system outputs "I am not sure" or gives multiple mutually contradictory answers, the observer receives a clear signal: "The readings here are unreliable; you need to judge for yourself." The pseudo-small fiber lowers this signal. It does not make uncertainty visible; rather, it covers over uncertainty with the **appearance** of certainty.

The pseudo-small fiber can arise from the observer's own cognitive bias, or be fostered by the optimization process of the observed system. When preference rewards give higher scores to the appearance of certainty, yet cannot effectively distinguish grounded from ungrounded certainty, and independent constraints sufficient to correct this bias are lacking, optimization may raise the expression of certainty without correspondingly reducing true uncertainty. Under these conditions, the system still cannot reliably determine the answer, but its visible output has been optimized to look "as if the fiber is small." This mechanism already has empirical literature support in AI alignment research—the review by Casper et al. (2023) points out that reward models in reinforcement learning from human feedback (RLHF) may misgeneralize into poor reward proxies—even when the training data are labeled correctly, a reward model may compute reward from unexpected, incidental features of the environment—and that reward models are fitted to what humans approve of rather than to what actually benefits them, so that the responses humans approve of are not necessarily genuinely beneficial—this provides a mechanism risk for the divergence of apparent accuracy from actual accuracy. The sycophancy study by Sharma et al. (2024) provides another relevant empirical illustration: models adjust the direction of their answers

based on the preferences implied by the user, which may be driven in part by preference judgments favoring sycophantic responses. These studies support the mechanism-level judgment that higher preference scores do not necessarily mean more reliable answers, but they do not directly measure the pseudo-small fiber as defined in this chapter—to incorporate them into pseudo-small fiber analysis, one must still specify how uncertainty is underestimated in the concrete task. In the language of the MCAF framework from the previous chapter, the pseudo-small fiber can be formulated as the masking of observable cognitive debt (D_c^{obs}) by narrative load (ϕ)—when misleading use accumulates while the observable signals still show normality, a gap opens between observable cognitive debt and true cognitive debt.

The Diagnostic Dilemma of the Pseudo-Small Fiber and Structural Inference. The misled observer, the diagnostician, and the time of observation must be distinguished. For an observer without effective calibration, the confident appearance of a single output is insufficient to determine reliably whether the true fiber is also small; this does not rule out later recognition of the earlier pseudo-small fiber through contradictions in outputs, prior knowledge, or external evidence. A change in perceived uncertainty after recognition does not remove the state that existed before recognition. However, the pseudo-small fiber can be **structurally inferred**: when the analyst identifies the structural conditions under which pseudo-small fibers are most likely to form (the optimization objective is driven only by preference signals, an independent calibration source is lacking, the observer passively receives readings), even if no single concrete instance of a pseudo-small fiber can be identified, there is reason to initiate preventive auditing against the risk of systematic production. When pseudo-small fibers are systematically produced at the systemic level by the optimization objective, the system is said to be in a state of **pseudo-calibration**.

Beyond verification aimed at specific outputs, the fiber framework also provides two indirect diagnostic paths:

- **Retrospective Diagnosis:** After a failure has occurred, the fiber framework provides a language to trace back "why everything appeared normal before the failure"—translating the problem of "not seeing" into a structural description of "the fiber was pseudo-shrunk," analogous to the "Swiss cheese model" in aviation accident investigation (Reason, 2000). A failure outcome alone does not warrant the inference that a pseudo-small fiber necessarily existed beforehand; the diagnosis must be checked against the evidence available at the time, the system's expression of certainty, and the user's judgment.
- **Structural Early Warning:** Even if no single concrete pseudo-small fiber can be detected, the fiber framework can annotate the structural conditions under which pseudo-small fibers **are most likely to form**: the optimization objective is driven only by preference signals, there is a lack of a true calibration source independent of preferences, and the observer passively receives readings with high verification costs. When these conditions are simultaneously satisfied, preventive measures at the institutional level (such as mandating the introduction of independent calibration sources, designing uncertainty-annotation mechanisms, or requiring pre-deployment red-team adversarial evaluation) are reasonable—even if no specific pseudo-small fiber has been detected. This is analogous to the **pre-emptive logic** of Failure Mode and Effects Analysis (FMEA) in engineering (International Electrotechnical Commission, 2018)—one does not need to wait for a bridge to collapse to know it could collapse; one can do a prospective diagnosis by analyzing material fatigue, load paths, and weld quality. Of course, what is borrowed here is FMEA's analytical sequence of identifying failure modes and their consequences before failure occurs, without comparing the precision of the two: what the fiber framework provides is a **structural identification of failure conditions**, not a specific failure probability.

Structural early warning is not a truth-value judgment about a specific output, but the **identification of system-level risk conditions**: when the optimization objective is decoupled from true calibration and the observer lacks independent verification channels, the system carries a risk of manufacturing false certainty. It must be emphasized: structural early warning concerns the conditions under which risk forms, not an assertion about the state of a specific system; concrete diagnosis must compare the evidence available at the time, the uncertainty disclosed by the system, and the judgment formed by the observer. Treating risk conditions under the precautionary principle (designing countermeasures as if pseudo-small fibers were present under these conditions) is a preventive decision, not a finding of fact.

Annotation of Assertion Types: The statements about "pseudo-small fibers" in this chapter and subsequent chapters can be divided into three types by their nature. To maintain diagnostic transparency, the following annotation symbols will be used where necessary in later text:

Symbol	Nature	Definition	Method of Verification
[S]	Structural Proposition	A conditional conclusion derived from definitions or known mathematical constraints under explicit premises; whether it applies still depends on whether the specific system satisfies these premises	Logical consistency check; seek counterexamples for falsification
[E]	Empirical Proposition	A factual assertion about the observable behavior of a specific system under specific conditions, supported by empirical literature or experimental data	Empirical testing (experiment, measurement, statistical inference)
[T]	Taxonomic Convention	The naming and categorization of phenomena within the framework, used to organize diagnostic language; not itself true or false	Check whether the definitions are consistent and the classification useful; subsuming a concrete case under the category still requires pertinent evidence

Institutional measures are normative claims; their goals, scope of application, and costs must be stated separately and are not to be co-labeled with structural propositions.

Examples: - "When reward-based discrimination cannot distinguish outputs with different true uncertainty but the same visible score, preference scores alone cannot establish that the true fiber has shrunk." **[S]** - "Sharma

et al. (2024) observed increases in feedback and mimicry sycophancy during the reinforcement-learning phase of Claude 2 fine-tuning." **[E]** - "We classify as a pseudo-small fiber the output-side phenomenon in which sycophantic behavior takes the form of a definitive tone that masks uncertainty." **[T]**

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§3 Structural Isomorphism with the "Clock Room"

The "Clock Room" thought experiment (Chapter 1) describes a scenario: a person is in a sealed room with N clocks before them, each displaying a different time. The room has no windows. Someone outside asks: "What time is it now?" The core proposition of the Clock Room is: under the condition that the mapping functions are completely unknown, they cannot answer.

This section does not attempt to "apply" the fiber framework to the Clock Room, or to "translate" the Clock Room into the language of fibers. Such a mapping across conceptual systems is itself lossy—the Clock Room has its own conceptual ecology (reading, calibration, time dimension, clock dimension), and the fiber framework has its own conceptual ecology (mapping function, fiber, visible/hidden subspace, pseudo-small fiber), and the concepts of each cannot be fully captured by the other. While acknowledging this lossiness, this section identifies the **structural resonance** between the two frameworks—the points at which they capture adjacent facets of the same deep structure.

§3.1 Resonance of Core Structures

This section does not attempt to establish a precise concept-to-concept translation between the two frameworks—this is impossible, because the respective conceptual ecologies of the Clock Room and the fiber framework are mutually irreducible. In identifying this structural resonance, the term "isomorphism" is used only to denote the correspondence in logical structure between the two, and does not claim conceptual inter-translatability.

The core structures of the two frameworks are isomorphic: the "reading" of the Clock Room corresponds to the "observed value" of the fiber framework; the "true time (T)" of the Clock Room corresponds to the "true state (T)" of the fiber framework; the "clock" of the Clock Room corresponds to the "mapping function (π)" of the fiber framework; the core proposition of the Clock Room—that under the condition of unknown mapping functions, any number of readings cannot uniquely determine the true time—corresponds to Proposition 1 of the fiber framework (the knowledge-conditioned identification set is the full set when the constraints on candidate mappings cannot exclude any true state).

The three layers of progressive irreversibility identified by the Clock Room—information deficit, dynamic corrosion, ontological dimension reduction—can be related to the three core propositions of the fiber framework: information deficit may take the form of an identification set that still contains multiple states—this may arise either from non-injectivity of the actual mapping or from insufficient knowledge of the mapping, and the two must be analyzed separately—and ontological dimension reduction corresponds to Proposition 3 (lost information cannot be recovered through post-processing alone). The second layer—dynamic corrosion—has a more indirect relation to Proposition 2: Proposition 2 states the possibility and conditions of calibration narrowing the identification set, whereas dynamic corrosion states the vulnerability of calibration to being eroded in use—the former is a positive capacity proposition, the latter a negative vulnerability proposition. The static formalization

of the fiber framework does not assign independent proposition status to dynamic corrosion; the tracking of this dimension is undertaken by the MCAF framework.

The two frameworks also have corresponding formulations on key dynamics. The Clock Room argues that "calibrating a clock with a clock is not calibration"—when the person in the room tries to infer the true time by comparing the readings of different clocks, they are only establishing relative relations among clocks, not the relation between any clock and the truth. In the language of fibers, this is "closed-loop pseudo-calibration": cross-validation within the system can produce internal consistency, but cannot guarantee alignment with the truth. Consistency is not equivalent to calibration.

The Clock Room also identifies an even more dangerous dynamic: when all clocks display exactly the same time, the person in the room is most likely to simply report that number, without the slightest hesitation. Yet their epistemological situation is exactly the same as when the clocks differ—they still have no window. What consistency most easily eliminates is not ignorance itself, but **the awareness of ignorance**. In the language of fibers, this is the Clock Room version of the pseudo-small fiber: all clocks displaying the same reading, the observer forming the perception of a small fiber, when in fact the fiber may still be large—all the clocks may share the same systematic bias.

§3.2 Neither Is Reducible to the Other

Despite these structural resonances, neither framework is reducible to the other. The Clock Room provides something the fiber framework does not: an ontological dimension—the distinction between the time dimension and the clock dimension. The Clock Room argues not only that "readings cannot uniquely determine the truth," but also that "dependence on readings changes the cognizer's state of being." When a person relies entirely on clock-face readings as the basis for decisions, they slip from the time dimension (a state of being in direct causal contact with the truth and able to cross-check it through multiple forms of perception and action) into the clock dimension (an operational state that only processes readings and lacks external cross-checking). The fiber framework is not concerned with ontology—it is concerned only with information structure.

Conversely, the fiber framework provides something the Clock Room does not: an operational set of analytical tools. The visible/hidden subspace decomposition, ordinal comparison of fiber sizes, the formal definition of the pseudo-small fiber—these tools allow the intuition of the Clock Room to be applied to concrete engineering scenarios. The Clock Room says "calibrating a clock with a clock is not calibration"; the fiber framework says "let us measure the size of the fiber inside this closed loop and see under what conditions it will inflate."

The relationship between the two frameworks is not that "one contains the other," but **complementarity**. The Clock Room provides the ontological ground for "why this matters"; the fiber framework provides the analytical tool for "how to operationalize it." The two resonate on the core structure of "readings are not the truth," but each goes in a different direction.

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§4 Conceptual Complementarity with the MCAF Framework

The MCAF framework established in the previous chapter is a cross-domain cognitive debt bookkeeping system. It uses the six-variable meta-formula—unlocked stock (C), media conductivity coefficient (Λ), narrative load

(ϕ), meta-audit capacity (A), human biological substrate (E_c), legitimacy base ($\|L\|$)—to model the projection-defense dynamic as an interdisciplinary chart of accounts, demarcating the structure of temporal asymmetry through the three clocks.

The relationship between the fiber framework and the MCAF framework is not that "the fiber is part of the MCAF framework" or that "the MCAF framework is an application of the fiber." They are two independent sets of analytical tools, each with its own conceptual ecology, working synergistically in the common direction of "making the invisible visible."

§4.1 Fiber as an Analytical Tool for the MCAF Framework

One of the core variables of the MCAF framework is narrative load (ϕ)—the narrative packaging and information asymmetry projected by the projecting party in pursuit of local optima. In this analysis, one function of narrative load is to alter the receiver's perception of the reading—making a large fiber look like a small fiber. In the language of the MCAF framework, narrative load eliminates the contrast conditions for auditing by blanketing the information environment; in the language of fibers, narrative load manufactures pseudo-small fibers by manipulating the formal features of the output that are easily perceived by the observer.

More specifically, the MCAF framework's "shadow debt"—the gap between true cognitive debt (D_c^{true}) and observable cognitive debt (D_c^{obs})—resonates analytically with the gap between the true fiber and the perceived fiber, but the two are not the same variable. Pseudo-small fibers can serve as one mechanism by which the consequences of debt remain unmanifested as observable cognitive debt for a long time; shadow debt can also arise from other delayed-manifestation mechanisms. The direction in which the true fiber changes on the hidden subspace requires independent testing and cannot be asserted a priori.

The MCAF framework's path of "compulsory bookkeeping"—making shadow debt visible, bringing observable cognitive debt closer to true cognitive debt—corresponds in the language of fibers to the operation of **annotating the fiber**: not eliminating the fiber, but honestly annotating the existence and size of the fiber. When a system is forced to disclose its range of uncertainty, known failure modes, and deviation characteristics, the observer's judgment about the uncertainty can move closer to the range supported by independent evidence, thereby weakening the pseudo-small fiber.

A Worked Example of Joint Analysis. Consider a hypothetical medical-advice scenario used for conceptual illustration: a user asks a large language model, "I've been feeling dizzy recently; what could be the cause?" The model answers in a fluent, definitive tone: "This is most likely caused by cervical spine issues; I suggest you get a cervical computed tomography (CT) scan."—The micro-anatomy of the fiber framework must first specify its object: the observation relation analyzed here maps potential causes to reported symptoms, rather than treating a list of causes as the complete inverse image of the model's answer. Suppose the reported symptoms remain compatible with multiple causes, while the model's single attribution and tone mislead the user into believing that the cause has essentially been determined. This constitutes an underestimation of substantial etiological uncertainty and can serve as a conceptual example of a pseudo-small fiber; it is not a diagnosis of a particular patient. The macro-tracking of the MCAF framework is as follows: if the user relies on this model for health advice over the long term, interactions that foster misinformed use and whose consequences have not yet manifested may accumulate shadow debt—the user's trust in the model may persist, while the consequences of misinformation have not yet fully manifested as observable cognitive debt (D_c^{obs}); the gap between true

cognitive debt (D_c^{true}) and its manifested portion may widen. When a certain instance of misinformation leads to a delayed medical consultation, the shadow debt undergoes liquidation. Disclaimers such as "please consult a medical professional" can serve as candidate interventions, but whether they reduce pseudo-small fiber risk depends on their content, presentation, and whether users adjust their judgments and actions accordingly; this requires testing for specific models, versions, and use settings. Even an effective intervention in one topic category does not establish that risks in other topics have been eliminated.

The two frameworks divide labor on the time scale: the fiber framework performs the "micro-anatomy" of a single interaction (How large is the possibility space of this answer? Does the definitive tone match the true uncertainty?), while the MCAF framework performs "macro-tracking" across periods (How have the unmanifested consequences associated with these uncertainties accumulated across periods? Who bears them? When may they be liquidated?). It must be noted: what the fiber framework outputs in this case is a qualitative judgment ("a pseudo-small fiber risk exists"), which the bookkeeping-discipline layer of the MCAF framework can record together with its consequences; developing scenario-specific quantitative bookkeeping further requires explicit quantitative indicators and scales. One possible research path is: first to specify the damage indicators, time scales, assignment rules for risk levels, and consequence weights in the scenario, and then to examine whether an explanatory quantitative mapping can be established. Before these steps are completed, high/medium/low risk can only be recorded jointly with consequence categories; the categories cannot be multiplied directly to obtain an estimated interval for the increment of shadow debt.[3] The synergy of the two frameworks is a conceptual coupling; their concrete quantitative interface—mapping from qualitative risk annotation to quantitative debt bookkeeping—requires subsequent empirical calibration.

§4.2 The MCAF Framework as the Bookkeeping Discipline for Fiber

Conversely, the MCAF framework provides a systematic "bookkeeping discipline" for the fiber framework. The fiber framework excels at analyzing the uncertainty structure of **single** observation scenarios—how large the fiber of this reading is, how the visible/hidden subspace is decomposed, whether a pseudo-small fiber exists. But it is not good at tracking how the consequences associated with these uncertainties accumulate over time, how they propagate in the system, and how they are systematically exploited by the projecting party.

The MCAF framework fills this gap. It integrates scattered fiber analyses into a cross-period, cross-domain cognitive debt tracking system using the compulsory questioning discipline of double-entry bookkeeping. The fiber framework reveals "how much possibility space lies behind this reading"; the MCAF framework tracks "how the unmanifested consequences associated with these uncertainties have accumulated across periods, who bears them, and when they may be liquidated."

The synergy of the two can be understood this way: the fiber framework provides the "microscope"—enabling the analyst to see clearly the information structure within a single mapping process; the MCAF framework provides the "balance sheet"—enabling the analyst to track the distribution and evolution of these information structures across the entire system. The two are complementary and cannot substitute for each other.

[3] This chapter does not provide the specific form of this quantitative mapping; this requires scenario-specific calibration in subsequent chapters. Here, only a demonstration of how the two frameworks synergistically pose questions is given, not an operational quantitative tool.

§4.3 Neither Is Reducible to the Other

The fiber framework is not the "micro-foundation" of the MCAF framework, nor is the MCAF framework the "macro-unfolding" of the fiber framework. The advantage of the fiber framework lies in **structural analysis**—dissecting the information loss and uncertainty distribution in a single mapping process; its core question is "in this observation, what is seen and what is hidden?" The advantage of the MCAF framework lies in **dynamic tracking**—tracking the accumulation, concealment, and liquidation processes of cognitive debt; its core question is "in this system, who is accumulating debt, when does the debt come due, and who pays the bill?" A complete analysis usually requires the synergistic operation of both, but the two are not the same conceptual system—they can simply be used in concert within the same analysis.

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§5 Resonance and Divergence with the Hermeneutic Circle Theory

Zhao Dingxin introduced the concept of the "hermeneutic circle" into sociological analysis to describe the epistemological situation in which social scientists find themselves when trying to understand social phenomena. The core structure of the hermeneutic circle is: the researcher can only ever start from partial information and attempt to understand the whole; the resulting theory is forever confined within a "hermeneutic circle," unable to fully escape the researcher's initial presuppositions and limited information. Zhao Dingxin (2022) points out that the process of theory construction in the social sciences is essentially a process of continuously "shrinking the hermeneutic circle"—by the five methods of mutual corroboration of macro-structure and micro-mechanisms, mutual corroboration of macro-structure, micro-mechanisms, and temporal sequences, ruling out alternative explanations, counterfactual reasoning, and increasing the amount of information about the problem being explained, the researcher can gradually approach a more reliable characterization of the whole, but can never close the hermeneutic circle itself.

§5.1 Structural Isomorphism and Divergence

A structural resonance exists between the two at the level of logical correspondence, not conceptual inter-translatability.[4] The 'partial information' of the hermeneutic circle corresponds to the 'observed value' of the fiber framework—the researcher can only ever access certain partial manifestations of the whole. The 'whole' corresponds to the 'true state'—the researcher's ultimate goal, but not directly accessible. The 'process of interpretation' corresponds to inference from observations to the underlying state, not to the mapping π that generates observations from that state; researchers' assumptions, methodological choices, and cognitive limitations may affect both the formation of evidence and the inference drawn from it. The comparison is limited to the epistemic structure in which partial information does not automatically determine the whole uniquely. In this book's retrospective rereading, the partial–whole tension marked by the hermeneutic circle can be compared with the range of compatible states marked by a fiber, but their 'sizes' cannot be directly equated. Zhao treats methodological shrinking as a way to improve empirical reliability, while also treating the true believer's 'shrinking to zero' as dogmatic closure; fiber size, by contrast, must be interpreted under a specified state space and measure.

[4] Zhao Dingxin's hermeneutic circle has dimensions specific to sociology (power, institutions, history), which the fiber framework does not address. The analogy between the two is limited to an illuminating similarity in the inverse-inference problem of "inferring the whole from partial information"; it does not imply conceptual inter-translatability.

A common danger identified by Zhao (2022), illustrated by the 'true believer'—'the hermeneutic circle going to zero' (author's translation)—is that the believer takes what he believes to explain everything. This book extends the warning to the false certainty of theory holders (a retrospective rereading)—the 'shrinking to zero' here is the interpreter's subjective conviction, not an evidence-based reduction of uncertainty—the researcher mistakenly believing that their theory has completely explained the phenomenon, forgetting that any theory is built upon specific partial information and theoretical presuppositions. The diagnosis developed further in this book is: **consistency is not calibration; confidence is not correctness.** In the hermeneutic circle, an internally coherent theory can completely deviate from the truth; in the fiber framework, an AI answer produced in a fluent, confident tone can be completely wrong. This book then extends the warning to visible performance and hidden risk under preference optimization: when a system performs increasingly 'well' on visible dimensions, it remains necessary to ask whether hidden dimensions obscured by this success narrative are accumulating deviations that the observer cannot see. This extension is an argument of the present book, not an AI claim already advanced in Zhao's original text.[5]

§5.2 Difference in Emphasis

Despite their structural resonance, the two frameworks differ in their diagnostic emphasis—not in opposition, but each illuminating different facets of the problem.

The hermeneutic circle emphasizes the observer side. Its core concern is the finitude of human cognitive capacity—the researcher can only ever start from the partial and can never exhaust all information about the whole. Shrinking the hermeneutic circle relies on strategies such as mutual corroboration, ruling out competing explanations, counterfactual reasoning, and increasing the amount of information about the problem being explained, rather than the mere accumulation of material or perspectives. Using a metaphor: the hermeneutic circle is concerned with the **limitations of the observer's perspective**—standing at different angles and seeing different sides, approaching the whole by increasing the number of observation angles.

The fiber framework emphasizes the mapping side. Its core concern is not the 'insufficient quantity' of information, but the fact that the mapping process from truth to observation may itself be a lossy transformation—even with a vast amount of information, if the mapping process distorts the information, the observer may end up farther from the truth. Using the same metaphor: the fiber framework is concerned with the **aberrations of the imaging system itself**—even if the observer stands at the correct angle, if the lens or reflective surface has systematic distortion, the observed picture will still deviate from the truth. Reward proxy failure in RLHF provides a conditional illustration: the issue is not merely the quantity of data or whether information is sufficient; even with abundant training material, if the reward signal is decoupled from genuine calibration, the model may still learn to output unreliable answers in a confident tone. In this illustration, reliable mapping structure cannot be inferred from data quantity alone, and reward proxy failure must not be conflated with information insufficiency as a single cause.

[5]This diagnosis can be compared, within limits, to a warning from Western philosophy. Heidegger in *Being and Time* distinguishes "beings" (Seiendes) from "Being" (Sein). Wang Defeng offers a concise Chinese formulation of this: "The principle of reason is to determine and classify the surrounding world using categories of thought—this is the determination of beings. In the determination of beings, Being is forgotten." (author's translation of Wang Defeng's lecture transcript, *The Ideal and Fate of Classical Art*; for the distinction between Being and beings, see also Heidegger, 1927/2006.) This chapter draws only a limited analogy between this warning and "mistaking readings for truth": both caution against treating a particular determination or representation as having exhausted its object. Heidegger's "Being," however, is not the unobserved part of a true-state space, and the distinction between beings and Being is not a set-theoretic division between observations and hidden states. The comparison concerns two cautions against the substitution of representation, not an identity of concepts.

This difference in emphasis leads to two different but complementary coping strategies. The strategy of the hermeneutic circle includes testing the competition among explanations, expanding the conditions of mutual corroboration, and increasing information; the fiber framework, within this common need, highlights the auditing of the observation mapping and its institutional conditions—checking whether the mapping process itself is distorted, annotating uncertainty, and when necessary introducing friction to block the propagation of pseudo-small fibers. The two strategies are not mutually exclusive, but simultaneously necessary: increasing perspectives is necessary, but when the mapping process of information is systematically distorted, relying solely on increasing perspectives may be counterproductive—repeatedly encountered biased information may keep reinforcing false certainty without adding discernment of the true state; whether it enlarges the identification set or aggravates the underestimation of uncertainty must be analyzed separately. Auditing the medium also requires new information, but this information is meta-information about **the mapping function itself**, not direct information about the observed object.

§5.3 Complementary, Not Substitutes

The hermeneutic circle theory and the fiber framework are not in a competitive relationship, but a complementary one. The hermeneutic circle theory reveals "the structural limitations of the cognizer in understanding the social whole"—its unit of analysis is the cognitive process of the human researcher. The fiber framework reveals "the structural limitations of any mapping system in transforming truth into observation"—its unit of analysis is the mapping process itself.

When a social scientist uses AI to assist research, the complementarity of the two frameworks becomes clear. The hermeneutic circle theory tells this researcher: the researcher can never exhaust all information about the object of study; theory is always built upon partial information. The fiber framework tells this researcher: the AI system used to assist the research is itself a lossy mapping—behind its output there is a possibility space that the researcher may not see; it may be manufacturing the illusion that "this is the answer" with the confident tone of a pseudo-small fiber. The two frameworks therefore operate synergistically—the hermeneutic circle theory calibrates the observer's perspective limitations; the fiber framework audits the aberrations of the mapping process itself.

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§6 Applicability Conditions and the Diagnostic Power Spectrum

The fiber framework is a diagnostic tool for structural risk, not a universal epistemological framework. Its diagnostic power is not uniform—it is strongest under certain conditions and significantly weakened under others. Clearly annotating these conditions is a requirement of the framework's own honesty. The "diagnostic power" in this section refers to the framework's irreplaceability and risk salience in the given scenario, not to the testability of concrete diagnostic conclusions—high-demand scenarios are often precisely those lacking independent evidence; structural risk annotation can stand while confirmation of concrete instances remains harder.

High Diagnostic Power Conditions (Scenarios where the fiber framework is most applicable):

- The observer passively receives readings and lacks the capacity to actively intervene in the true state (e.g., financial market participants facing price signals, the public facing algorithmic recommendations);

- The system's optimization objective is primarily driven by preference signals rather than independently verifiable true standards (e.g., AI systems optimized by RLHF, bureaucratic systems assessed by key performance indicators (KPIs));
- There is a lack of an independent external verification channel separate from the original mapping function (no "window" to see the truth);
- The mapping chain is relatively long, passing through multiple layers of intermediation, and the risk that the intermediaries share a systematic bias is high.

These conditions describe **cognitive environments where calibration mechanisms are absent**—closed bureaucratic systems, manipulated information environments, certain AI deployment scenarios. The fiber framework is positioned precisely as a diagnostic tool for structural risks in such environments.

Low Diagnostic Power Conditions (Where the applicability of the fiber framework needs careful assessment):

- **When the mapping function is sufficiently known.** If the properties of the mapping function are known, and the knowledge-conditioned identification set or the task-relevant uncertainty interval is already sufficiently narrow, the diagnostic value is limited. The problem then shifts from "how large is the fiber?" to "is the fiber small enough to meet the requirements of the current task?"
- **When the observer can actively intervene in the true state.** The essence of the experimental method is to actively choose inputs and observe changes in outputs, thereby acquiring information about the mapping function. Experiments can narrow the identification set by adding observational constraints, or form a joint observation mapping with a smaller inverse image.
- **When the task has an independent external verification channel.** Code generation can be compiled and tested by running it. Mathematical reasoning can be formally verified. Scientific predictions can be checked against experiments. In these scenarios, external verification provides an independent channel that can constrain optimization and use; only when verification procedures are actually incorporated into the evaluation and use pipeline, and the object of verification matches the task at hand, are the conditions for forming a pseudo-small fiber actually weakened. This condition constitutes the natural applicability boundary for the "algorithmic *xiangyuan*" framework in Part II (AI Engineering Diagnostics) of this book (cf. the discussion of the constraint source outside the discriminative function d in Chapter 10 §5).
- **When the observer maintains the habit of active calibration.** (Note: this item describes a behavioral characteristic of the observer, not a scenario characteristic, and its nature differs from the previous three.) An observer who actively seeks independent information sources, cross-verifies across sources, and defers judgment will not easily have their perceived fiber converge toward a pseudo-small fiber. Condition four is less a boundary of the framework and more a defensive measure against pseudo-small fibers—it describes the path by which an observer actively breaks the conditions for forming pseudo-small fibers.

These two sets of conditions are not a binary opposition, but a spectrum: from completely passive observation, purely preference-driven, zero external verification (highest audit demand) to completely active experimentation, independently verifiable, active calibration (lowest audit demand). Most real-world scenarios lie somewhere on this spectrum. The analyst should identify the position of the current scenario on the spectrum and adjust the application intensity of the fiber framework according to the strength of diagnostic power.

If a scenario does not meet the above high diagnostic power conditions, forcibly applying the fiber framework

may manufacture a spurious uncertainty problem—describing a problem that could be solved through experimentation as a structural uncertainty predicament. This is the risk of framework misuse and requires active prevention.

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§7 Conclusion: Annotating the Fiber

The core proposition of the fiber framework can be summarized in one sentence: **Between any observation and the truth, there always lies a possibility space—the fiber.** For a non-singleton inverse image, the reading and its subsequent processing alone cannot eliminate the indistinguishability among those states: when the mapping is lossy, there exist readings that correspond to multiple true states; when no constraint on the candidate mappings suffices to rule out any true state, the knowledge-conditioned identification set is the full set; when preference rewards favor the appearance of certainty and true calibration is insufficient to constrain this tendency, optimization may manufacture pseudo-small fibers without correspondingly reducing true uncertainty.

These propositions each have scholarly precursors. Representational measurement theory understands measurement as the construction of homomorphisms from empirical relational structures to numerical relational structures and requires clarification of their assumptions (Krantz et al., 1971). Partial identification reports the parameter values compatible with the observed distribution and adopted assumptions rather than passing off partial identification as point identification (Manski, 2003). Information theory uses conditional entropy, mutual information, and the data processing inequality to characterize the informational boundaries of observation and post-processing (Shannon, 1948). The holist warning about theory testing holds that empirical tests typically involve both theories and auxiliary hypotheses, so the same observational evidence can be compatible with different theories (Quine, 1951). The Law of Requisite Variety characterizes limits on regulatory capacity: for a given system and regulatory objective, complete regulation is impossible if the effective disturbances that still require distinct treatment exceed the regulator's response variety; disturbances already attenuated or merged by the system must be accounted for separately (Ashby, 1956). The contribution of the fiber framework lies not in inventing these insights, but in providing a unified intuitive language and cross-domain conceptual anchor for them. The reader does not need any mathematical background to understand that 'one shadow can correspond to multiple objects.' Once this intuition is grasped, it can be applied to any scenario where a mapping relationship exists.

The practical direction of the fiber framework is not "eliminating the fiber"—for a non-singleton inverse image, the reading and its subsequent processing alone cannot eliminate the indistinguishability among those states. Its practical direction is **annotating the fiber**: beside every reading, honestly annotate "how large is the possibility space behind this reading," "what factors are ignored by this mapping," "does the definitive tone of this system match its true uncertainty." Annotating the fiber does not eliminate uncertainty, but it makes uncertainty visible. For decision-makers relying on that reading, being aware of the boundaries of possibility is usually safer than mistakenly believing that the reading uniquely determines the underlying true state.

Operational Guide: A Question Checklist for Fiber Diagnosis. Even if the absolute size of the fiber cannot be measured, the following questions can activate the diagnostic intuition of the fiber framework:

- How long is the mapping chain of this reading? How many layers of intermediation does it pass through?

- Is there an independent calibration point in the mapping process? When was the last calibration?
- If this reading is wrong, how long would it take to be discovered? Does the output tone of the system match its verifiable accuracy rate?
- Who bears the consequences of a possible error in this reading—the information receiver, or the system's manufacturer?

These questions do not provide numerical answers, but they change the analyst's relationship to the reading—from passive reception to active questioning.

Possible Paths from Concept to Operation. The fiber framework is currently at the stage of a conceptual language. Its transformation from concept to operation requires advancement in at least three directions:

1. **Development of Proxy Measures:** Develop observable proxy indicators when the true fiber size cannot be measured directly—for example, the consistency of the output distribution, the stability of the model under adversarial perturbations, and the degree of divergence among different independent calibration sources. These proxy indicators do not directly equal fiber size; whether they correlate with a clearly defined uncertainty measure in a particular setting, and whether the direction of that correlation is stable, requires independent empirical calibration. It should be noted that the interpretation of proxy indicators is not monotonic: for example, high output consistency could reflect genuinely low uncertainty (the fiber is indeed small), or it could reflect the system overfitting to a certain output pattern (pseudo-small fiber). Therefore, proxy indicators need to be used in conjunction with an independent calibration baseline, rather than serving as a measure of fiber size on their own. Chapter 13 §8 has already provided three candidate proxy indicators for the hidden subspace (out-of-distribution performance decay rate, tail-group performance disparity, effective dimensionality of the model's internal representations) as a first-step attempt in this direction.
2. **Construction of Calibration Baselines:** For specific domains, establish external calibration baselines independent of the system's training data. The value of a baseline is not "absolute truth," but providing a reference frame independent of the system being evaluated—a "window."
3. **Design of Institutional Audit Procedures:** Translate the fiber framework's risk annotations into institutional audit practices—for example, requiring high-risk AI systems to simultaneously output uncertainty estimates when outputting definitive answers, or mandating pre-deployment red-team adversarial evaluation as a precondition.

These directions lie beyond the scope of this chapter, but they mark the possible paths for the fiber framework to move from a 'diagnostic language' to an 'operational tool.' Each of these three paths is itself a complex research program. At present, the framework only provides the argument for 'why these paths are needed,' not a roadmap for 'how to carry them out.'

Whether Proxy Measures Can Stand on Their Own. The development of the above proxy indicators faces a circular dilemma—in open scenarios where the target variable (true fiber size) cannot be directly measured, proxy indicators cannot obtain conclusive verification; in environments where the state space and mapping are controllable, the proxy relationship and its scope of applicability can be tested. The framework's response to this is not to claim to have resolved this dilemma, but to reposition the role of the framework as a **testable coordinate for discussion** rather than a **confirmed theory**: definitions and taxonomic conventions are assessed by clarity

and utility, conditional propositions by whether their premises are met, and empirical judgments are converted into testable questions through proxy indicators. The function of proxy indicators is not to provide a precise estimate of fiber size, but to anchor the previously unobservable debate ("How large is the hidden subspace of this system?") onto observable quantities ("What is its out-of-distribution decay rate?"). A critic may argue that the choice of proxy indicators is inappropriate, but this argument itself is already more operational than a purely conceptual discussion of the "hidden subspace." The V1.0 of the framework (versioning defined in Chapter 13 §8.2) honestly annotates its limits here; one direction for the evolution of V2.0 is to establish a statistical calibration relationship between proxy indicators and true fiber size in a controlled environment.

This is precisely where the fiber framework converges with the "Clock Room," the MCAF framework, and the hermeneutic circle theory. The "Clock Room" says: **This is not time. This is a clock.** The fiber framework says: **Behind the reading of this clock, there is a possibility space that the observer may not see.** The MCAF framework says: **How have the unmanifested consequences associated with these uncertainties accumulated across periods? Let us put it on the books.** The hermeneutic circle theory says: **Any theory is built upon partial information; the hermeneutic circle can only be shrunk, never eliminated.**

All four work in the same direction: **making the invisible visible.**

The next chapter will shift from information structure to ontology: what is eroded by the clock is not merely an information-processing capacity, but also a state of being—psychosocial homeostasis in the sense of Francis L. K. Hsu. The fiber framework tells us "what is hidden"; the next chapter presses the question "what does that which is hidden mean for the human being?"

How Subsequent Chapters Will Use This Chapter. The fiber framework established in this chapter will repeatedly appear in subsequent chapters as a public language for diagnostic intuition, but not as a precise measurement instrument; rather, as a system for naming and identifying structural risks: Part II (AI Engineering Diagnostics) will use the fiber framework to identify pseudo-small fibers in scenarios such as algorithmic *xiangyuan* and RLHF failure; Part III (Financial Market Analysis) will use the fiber framework to dissect the information loss in financial mappings such as price signals and risk models. The reader does not need to memorize all the formalized content of this chapter, but only needs to take away one intuition: **in observations involving information loss, the same reading may correspond to more than one true state.**

The Unfolding of the Framework in Specific Domains. This chapter has established public concepts such as fiber, visible and hidden subspaces, pseudo-small fiber, and the gradient of detectability. §1 discusses the information loss caused by mapping; §2.2 distinguishes visible from hidden dimensions; §2.4 defines the pseudo-small fiber; §3.1 discusses the structural isomorphism with the Clock Room and closed-loop calibration; §4 explains the complementary relation with the MCAF framework.

How these concepts are organized into diagnostic layers must be worked out in connection with concrete objects: Chapter 13 unfolds the "4+1+3" structure in AI scenarios, while Chapter 17, in turn, uses the five-layer progressive structure of the fiber framework (from L0 mapping-chain structure to L4 incentive capture) to diagnose how the financial clock slides from "pointing to the truth" to "self-reference." The two share the same analytical concepts, but identical layer numbers do not mean identical domain meanings; incentive capture (L4) in the financial part is not the same as narrative load multiplication (L4) in the AI part.

The following presents the unfolding in the AI part. The four diagnostic layers check different questions in

sequence; the listing does not assert that all systems already exhibit four kinds of failure:

- **L0 Mapping Chain Structure:** Examine the intermediaries between the truth and the reading, and the information lost—"How many layers lie between the truth and the reading? What does each layer discard?"
- **L1 Optimization Dynamics:** Examine what the evaluation objective covers and omits, and what constraints the hidden dimensions are subject to. Improvement in visible performance is not by itself sufficient to establish that hidden deviations have been eliminated—"Where is the optimization doing well, and where does it lack effective constraint?"
- **L2 Pseudo-Calibration:** Examine the disconnection between a true fiber that remains large and a perceived fiber compressed small—"Does the appearance of certainty mask a substance that remains uncertain?" The alignment paradox is one of its concrete forms within the preference-fitting paradigm: when reward-based discrimination cannot distinguish genuine alignment from *xiangyuan* on the hidden subspace, the two are indistinguishable in the direction of score optimization, and a higher preference score does not by itself certify genuine alignment (Chapter 10 §5).
- **L3 Multi-Mapping Recursion:** Track how the output of one mapping becomes the input of subsequent mappings, distinguishing one-way cascades from feedback that returns into inputs, evaluation, or training. The examination focuses on whether provenance, applicability conditions, and uncertainty are preserved, and whether independent calibration can intervene.

One emergent layer examines the system's interaction with humans and its discursive conditions:

- **L4 Narrative Load Multiplication:** Examine how interaction and narrative jointly alter the conditions for human calibration. Along one implementation path analyzed in Chapter 12, continuous delegation reduces the nodes of active judgment, reliable performance within the visible range masks relevant limits, and effective challenge is reduced to avoid user discomfort; when these mechanisms act on the same user, the calibration space is synergistically compressed. "Multiplication" marks this combination of mechanisms, not numerical multiplication. Whether it further produces a net decay of discriminative capacity must be judged together with duration and with independent calibration and sources of recovery (Chapter 12 §2, §5). Chapter 13 §6 examines its deployed implementations and actual performance, not substituting asymptotic direction for empirical judgment over finite horizons.

Three transversal dimensions cross-cut the diagnostic layers above; each is detailed in subsequent chapters:

- **Political Economy Dimension:** The ownership of the mapping function, the property rights structure of data, the institutional arrangements of auditing power.
- **Physical Substrate Dimension:** Computing power, energy consumption, planetary boundaries—the material conditions of mapping systems.
- **Temporal Dynamics Dimension:** The accumulation, liquidation, and temporal asymmetry of cognitive debt—the core object tracked by the MCAF framework.

This sequence moves from how readings are produced, to which performances are rewarded and how certainty is presented, to how readings enter subsequent judgment. Optimization takes concrete mappings and evaluation objectives as its object; when it systematically produces pseudo-small fibers, this constitutes the pseudo-calibration of §2.4. Multi-mapping relations do not presuppose pseudo-calibration; but when pseudo-calibrated outputs are taken by subsequent links as already-verified grounds, their deviations—together with their ap-

pearance of certainty—may be inherited and reinforced through feedback. This is the linkage of conditional mechanisms, not a sequence of four failures that every system undergoes in turn.

This "4+1+3" structure is one operational unfolding of the public concepts in AI scenarios; it neither replaces this chapter's definitions nor prescribes identical layer meanings for the financial part. Readers may regard this chapter as a conceptual vocabulary and the domain diagnoses as ways of applying these concepts to concrete objects; specific deployment prototypes and criteria are given in the respective chapters.

For the academic genealogy, parallel traditions, and scholarly positioning of the fiber framework, see Chapter 4 Appendix A: "A Scholarly Review of the Fiber Framework."

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